Problem Statement : Write a JavaScript program for performing Arithmetic Operations.

HTML CODE :

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <title>Arithmetic Operations</title>
   <link rel="stylesheet" href="calculator.css">
</head>
<body>
   <div class="calc">
      <h1>Arithmetic Operations</h1>
      <p class="sub">Enter two numbers and pick an operation</p>

      <div class="fields">
         <div>
            <label for="num1">First number</label>
            <input type="number" id="num1" placeholder="e.g. 12">
         </div>
         <div>
            <label for="num2">Second number</label>
            <input type="number" id="num2" placeholder="e.g. 4">
         </div>
      </div>

      <div class="ops">
         <button type="button" onclick="calculate('+')">Add +</button>
         <button type="button" onclick="calculate('-')">Subtract &minus;</button>
         <button type="button" onclick="calculate('*')">Multiply &times;</button>
         <button type="button" onclick="calculate('/')">Divide &divide;</button>
         <button type="button" onclick="calculate('%')">Modulus %</button>
         <button type="button" class="clear" onclick="clearAll()">Clear</button>
      </div>

      <div class="screen" id="screen">0</div>
   </div>

   <script>
      var symbols = { "+": "+", "-": "-", "*": "\u00D7", "/": "\u00F7", "%": "%" };

      function showResult(text, isError) {
         var screen = document.getElementById("screen");
         screen.innerHTML = text;
         screen.className = isError ? "screen error" : "screen";
      }

      function calculate(op) {
         var a = parseFloat(document.getElementById("num1").value);
         var b = parseFloat(document.getElementById("num2").value);

         if (isNaN(a) || isNaN(b)) {
```

```
            showResult("Please enter both numbers.", true);
            return;
        }

        var answer;
        if (op === "+") {
            answer = a + b;
        } else if (op === "-") {
            answer = a - b;
        } else if (op === "*") {
            answer = a * b;
        } else if (op === "/") {
            if (b === 0) {
                showResult("Cannot divide by zero.", true);
                return;
            }
            answer = a / b;
        } else if (op === "%") {
            if (b === 0) {
                showResult("Cannot find modulus with zero.", true);
                return;
            }
            answer = a % b;
        }

        answer = parseFloat(answer.toFixed(10));
        showResult(a + " " + symbols[op] + " " + b + " = " + answer, false);
    }

    function clearAll() {
        document.getElementById("num1").value = "";
        document.getElementById("num2").value = "";
        showResult("0", false);
    }
  </script>
</body>
</html>
```

CSS CODE :

```
:root {
  --navy: #14213d;
  --teal: #2ec4b6;
  --sand: #fdf6e3;
  --amber: #ffb703;
  --screen: #0b132b;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  padding: 16px;
  display: flex;
  justify-content: center;
  align-items: center;
  background: linear-gradient(160deg, #14213d, #1f3a5f);
  font-family: Arial, Helvetica, sans-serif;
}

.calc {
  width: 100%;
  max-width: 460px;
  background-color: var(--sand);
```

```
    border-top: 8px solid var(--amber);
    border-radius: 20px;
    padding: 28px;
    box-shadow: 0 10px 30px rgba(0, 0, 0, 0.45);
}

h1 {
    font-family: Georgia, 'Times New Roman', serif;
    text-align: center;
    color: var(--navy);
    margin: 0 0 4px 0;
}

.sub {
    text-align: center;
    color: #5c677d;
    margin: 0 0 20px 0;
}

.fields {
    display: grid;
    gap: 14px;
    margin-bottom: 18px;
}

label {
    font-weight: bold;
    color: var(--navy);
}

input[type="number"] {
    width: 100%;
    margin-top: 6px;
    padding: 12px;
    font-size: 1.1rem;
    border: 2px solid #b8c1cc;
    border-radius: 10px;
}

input[type="number"]:hover {
    border-color: var(--navy);
}

input[type="number"]:focus {
    border-color: var(--teal);
    outline: none;
}

.ops {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(120px, 1fr));
    gap: 10px;
}

.ops button {
    background-color: var(--navy);
    color: #ffffff;
    border: none;
    border-radius: 12px;
    padding: 14px;
    font-size: 1rem;
    cursor: pointer;
    transition: background-color 0.2s, transform 0.2s;
}

.ops button:hover {
    background-color: var(--teal);
```

```
    color: var(--navy);
    transform: translateY(-2px);
}

.ops button.clear {
    background-color: var(--amber);
    color: var(--navy);
}

.ops button.clear:hover {
    background-color: #ffd166;
}

.screen {
    background-color: var(--screen);
    color: var(--teal);
    font-family: 'Courier New', Courier, monospace;
    font-size: 1.4rem;
    text-align: right;
    min-height: 72px;
    margin-top: 18px;
    padding: 18px;
    border-radius: 12px;
    word-wrap: break-word;
}

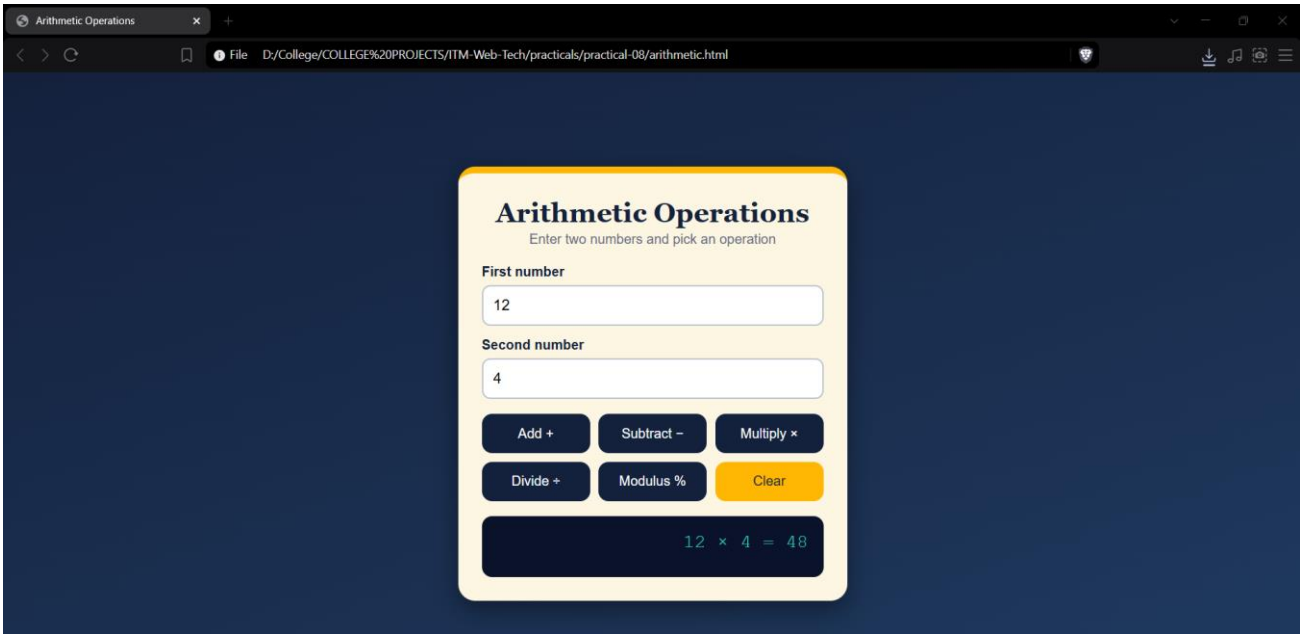
.screen.error {
    color: #ff6b6b;
    font-size: 1.1rem;
}

@media (max-width: 480px) {
    .calc {
        padding: 20px;
    }
}

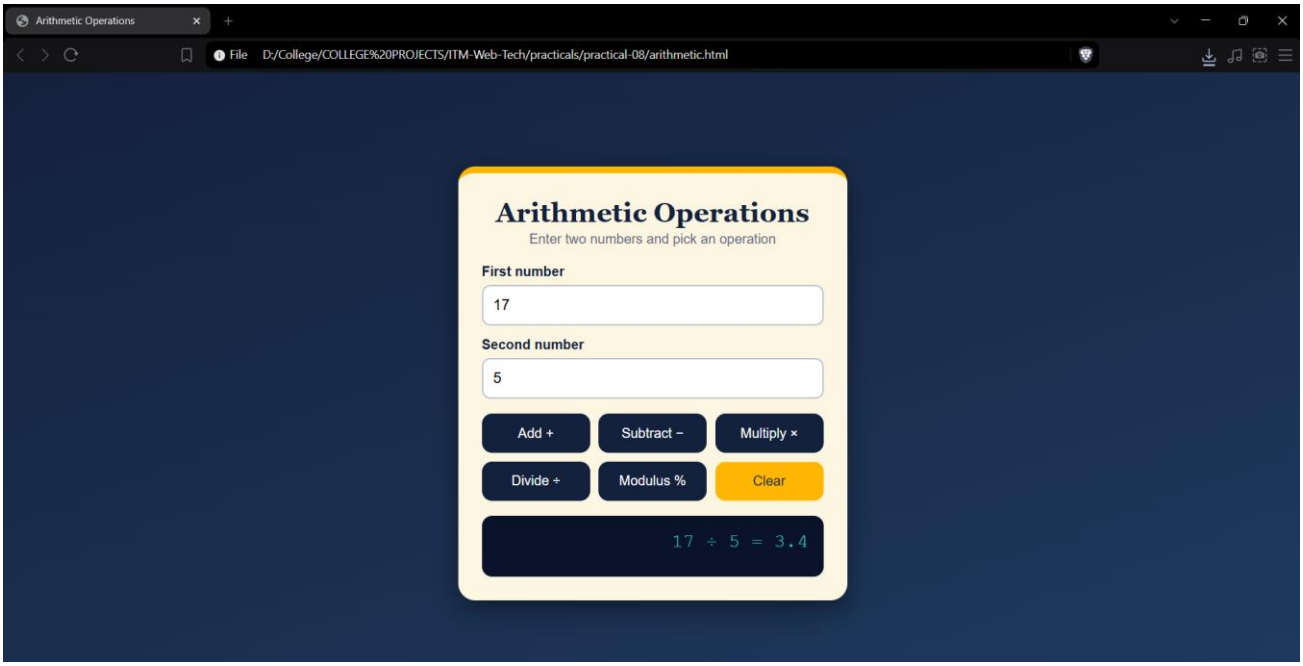
@media (prefers-reduced-motion: reduce) {
    .ops button,
    .ops button:hover {
        transition:  none;
        transform: none;
    }
}
```

OUTPUT :

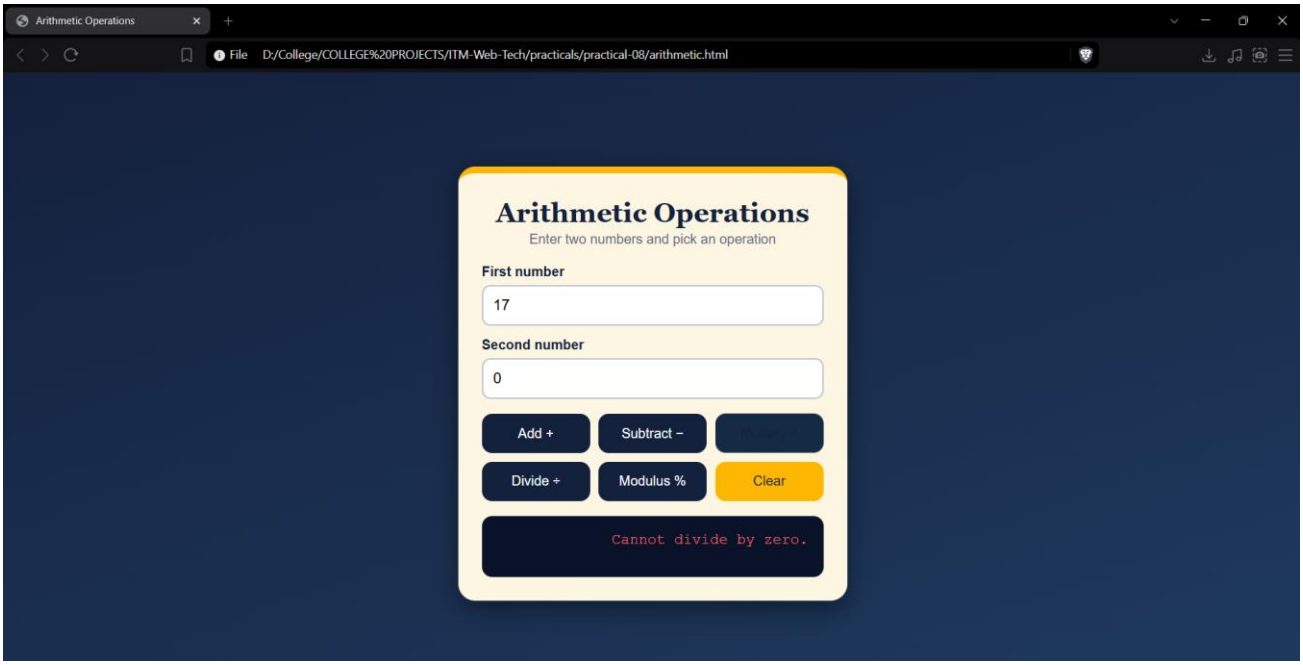
Multiplication (12 x 4)



Division (17 / 5)



Division by zero



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